

Technical Appendix: Higher Criticism for Adaptive Retrieval

Anonymous submission

Abstract

This supplementary document collects material referenced in the main paper but held out for space: the project roadmap, the full comparison-methods table, the CrossEntityQA construction, and the null-distribution uniformity diagnostics.

1 Project Roadmap

Wk 4 — Foundations. Implemented the HC module (p -values from a per-query null, γ guardrail) and the indexing stack on the CRAG (Meta AI 2024) corpus (recursive chunking, 384/50 overlap; OpenRouter LLM interface). *Direction change:* a k -sweep with `all-mpnet-base-v2` favoured $k=10$, but BGE (Xiao et al. 2023) gave a clear recall lift on a 1K-doc subset, so BGE became the default embedder.

Wk 5 — Survey of adaptive retrievers. Catalogued competing approaches (Reciprocal Rank Fusion, CAR (Xu et al. 2025), uncertainty filtering) to position HC as a global, distribution-aware test, and fixed the comparison set (Bonferroni (Dunn 1961), Benjamini-Hochberg (Benjamini and Hochberg 1995), gap-based Adaptive-K (Taguchi, Maekawa, and Bhutani 2025)). First HC vs. Top-K runs.

Wk 6 — Long-context regime: HoloBench. First HC vs. gap-based Adaptive-K on HoloBench (Maekawa, Iso, and Bhutani 2024). Judge scores tracked closely; *iteration:* diagnosed as a generation/judge bottleneck on aggregation, not a retrieval tie, which motivated stronger retrieval-only metrics.

Wk 7 — Multi-hop regime + dataset-aware γ . Repeated on HotpotQA (Yang et al. 2018). *Change in direction:* a stark dense-vs-sparse asymmetry meant the same γ behaved very differently, so γ was treated as dataset-aware. Began Adaptive-RAG (Jeong et al. 2024) and FlashRAG (Jin et al. 2025) integration.

Wk 8 — Pipeline integrations + FiQA. Plugged HC into Adaptive-RAG as a fourth strategy; on HotpotQA it beat single-step ONE-R and lost to multi-step IRCOT, as expected. Registered HC and Berk-Jones (Berk and Jones 1979) as FlashRAG post-processors on BEIR SciFact (Thakur et al. 2021; Wadden et al. 2020), FiQA (Thakur et al. 2021), and MS MARCO. *Iteration:* FiQA was the first single-hop negative result and triggered the position study behind the scattered-relevance analysis.

Wk 9 — CrossEntityQA construction. *Change in direction:* rather than keep tuning on off-the-shelf data, we built a sparse-relevance benchmark to surface HC’s target regime (SPARQL Wikidata clusters $\rightarrow$ Wikipedia passages $\rightarrow$ LLM queries answerable

This is an anonymized submission for review purposes only. Distribution, citation, or public sharing of this manuscript is strictly prohibited. Copyright and publication details will appear in the final version if accepted.

only by aggregating several passages; $\sim 2\%$ relevance, controllable K).

Wk 10 — Negative result: CrossEntityQA + BM25. HC *lost* to fixed Top-5 under BM25, whose score magnitude encodes lexical match, not the categorical constraints these queries impose. *Change in direction:* dropped BM25 for CrossEntityQA and committed to a dense retriever.

Wk 11 — Final breadth head-to-head. With the dense pipeline HC won on passage F1 on CrossEntityQA and was competitive with single-step retrieval on MuSiQue (Trivedi et al. 2022), closing the breadth picture and motivating the controlled Amazon study.

2 Comparison Methods

Table 1 lists the selection rules compared in the study. “Concept” is the rule that turns a candidate pool into a document set; the statistical methods share HC’s p -value machinery and differ only in the decision rule.

3 CrossEntityQA Construction

Figure 1 shows the CrossEntityQA construction pipeline. It guarantees a controllable per-query relevance count ($K=2-20$) and a sparse corpus (about 2% relevant), the two properties that make the benchmark a stress test for adaptive- k retrieval.



Figure 1: CrossEntityQA construction. Wikidata entity groups (WDQS/SPARQL) are split into constraint subclusters with a target relevance count $K=2-20$; a Wikipedia passage library supplies the texts; the builder generates LLM queries, filters for quality, and controls negatives and sparsity, exporting corpus, queries, and qrels.

4 Null-Distribution Diagnostics

This appendix verifies that the p -values are uniform under the null. On BEIR FiQA, under the *empirical* null the non-relevant p -values are uniform (KS $p=0.13$; 5.0% below 0.05), whereas a *parametric normal* null is not (KS $p=0.000$, an excess spike near zero), which is why we keep the empirical

Method	Concept	Reference
Fixed Top- k	rank cutoff	—
Adaptive-K	largest score gap	(Taguchi, Maekawa, and Bhutani 2025)
Adaptive-RAG	complexity routing	(Jeong et al. 2024)
FlashRAG post-proc.	cutoff plug-in	(Jin et al. 2025)
Full-context	all chunks to the LLM	—
Self-route	retrieve + routing prompt	—
HC (ours)	Higher Criticism statistic	(Donoho and Jin 2004)
Berk-Jones	goodness-of-fit statistic	(Berk and Jones 1979)
Bonferroni	FWER threshold	(Dunn 1961)
Benjamini-Hochberg	FDR step-up	(Benjamini and Hochberg 1995)
CAR	cluster cutoff	(Xu et al. 2025)
Per-query recal. (ours)	rescale z -scores to the global null	—

Table 1: Methods compared. All share HC’s p -value machinery except the fixed and cluster rules.

null; relevant documents carry systematically small p -values (Cohen’s $d = -1.33$), and the QQ plot of empirical p -values sits on the uniform diagonal (Figure 2).

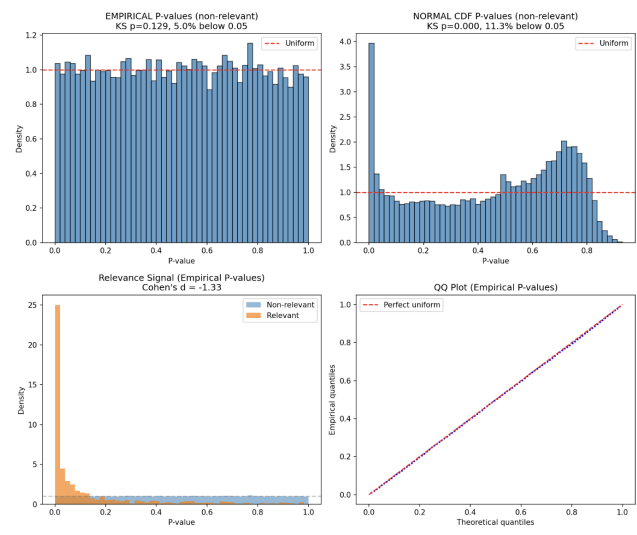


Figure 2: p -value uniformity diagnostics under the null (BEIR FiQA). The empirical null is uniform (top-left; QQ bottom-right) while a parametric normal null is not (top-right); relevant documents separate in p -value space (bottom-left).

References

Benjamini, Y.; and Hochberg, Y. 1995. Controlling the False Discovery Rate: A Practical and Powerful Approach to Multiple Testing. *Journal of the Royal Statistical Society. Series B (Methodological)*, 57(1): 289–300.

Berk, R. H.; and Jones, D. H. 1979. Goodness-of-Fit Test Statistics that Dominate the Kolmogorov Statistics. *Zeitschrift für Wahrscheinlichkeitstheorie und Verwandte Gebiete*, 47(1): 47–59.

Donoho, D.; and Jin, J. 2004. Higher Criticism for Detecting Sparse Heterogeneous Mixtures. *The Annals of Statistics*, 32(3): 962–994.

Dunn, O. J. 1961. Multiple Comparisons Among Means. *Journal of the American Statistical Association*, 56(293): 52–64.

Jeong, S.; Baek, J.; Cho, S.; Hwang, S. J.; and Park, J. C. 2024. Adaptive-RAG: Learning to Adapt Retrieval-Augmented Large Language Models through Question Complexity. In *Proceedings of NAACL*.

Jin, J.; Zhu, Y.; Yang, X.; Zhang, C.; and Dou, Z. 2025. FlashRAG: A Modular Toolkit for Efficient Retrieval-Augmented Generation Research. In *Proceedings of the ACM Web Conference (WWW)*.

Maekawa, S.; Iso, H.; and Bhutani, N. 2024. Holistic Reasoning with Long-Context LMs: A Benchmark for Database Operations on Massive Textual Data. In *Proceedings of NeurIPS Datasets and Benchmarks*.

Meta AI. 2024. CRAG: Comprehensive RAG Benchmark. <https://github.com/facebookresearch/CRAG>.

Taguchi, C.; Maekawa, S.; and Bhutani, N. 2025. Efficient Context Selection for Long-Context QA: No Tuning, No Iteration, Just Adaptive-k. In *Proceedings of EMNLP*. Megagon Labs.

Thakur, N.; Reimers, N.; Rücklé, A.; Srivastava, A.; and Gurevych, I. 2021. BEIR: A Heterogeneous Benchmark for Zero-shot Evaluation of Information Retrieval Models. In *Proceedings of NeurIPS Datasets and Benchmarks*.

Trivedi, H.; Balasubramanian, N.; Khot, T.; and Sabharwal, A. 2022. MuSiQue: Multihop Questions via Single-hop Question Composition. *Transactions of the Association for Computational Linguistics*, 10: 539–554.

Wadden, D.; Lin, S.; Lo, K.; Wang, L. L.; van Zuylen, M.; Cohan, A.; and Hajishirzi, H. 2020. Fact or Fiction: Verifying Scientific Claims. *Proceedings of EMNLP*.

Xiao, S.; Liu, Z.; Zhang, P.; and Muennighoff, N. 2023. C-Pack: Packaged Resources to Advance General Chinese Embedding. <https://huggingface.co/BAAI/bge-base-en-v1.5>.

Xu, Y.; Gupta, V.; Aggarwal, R.; Mahadevan, V.; and Krishnamachari, B. 2025. Cluster-based Adaptive Retrieval: Dynamic Context Selection for RAG Applications. arXiv:2511.14769.

Yang, Z.; Qi, P.; Zhang, S.; Bengio, Y.; Cohen, W. W.; Salakhutdinov, R.; and Manning, C. D. 2018. HotpotQA: A Dataset for Diverse, Explainable Multi-hop Question Answering. In *Proceedings of EMNLP*.